

Q.5 In radars which device allows the use of same antenna for both transmission and reception

- a) Duplexer b) Converter
- c) Mixer d) Oscillator

Q.6 A Radar is used for

- a) Ground mapping b) Airports surveillance
- c) Weather forecast d) All of above

Q.7 Which of the following is used to join waveguides of different dimensions

- a) Twist b) Circulator
- c) Isolator d) Taper

Q.8 Radiation pattern of Horn antenna is

- a) Unidirectional b) Bidirectional
- c) Omnidirectional d) None

Q.9 E layer is at a height of

- a) 50-100 Km b) 100-140 Km
- c) 140-250 Km d) 250-400 Km

Q.10 Frequency range of S band is

- a) 2-4 GHz b) 4-8 GHz
- c) 8-12 GHz d) 1-2 GHz

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 The dominant mode is rectangular waveguide is _____.

Q.12 Expand the term IMPATT.

Q.13 Define Cut off wavelength.

Q.14 PPI stands for _____.

Q.15 What is the use of a microwave bend?

Q.16 Frequency range of UHF band is _____.

Q.17 Microwave tubes work on the principle of _____ modulation.

Q.18 A magic TEE has _____ no of ports.

Q.19 Expand FMCW radar.

Q.20 Define Directivity.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Define free space wavelength, how it is related with guided wave length?

Q.22 Why TEM mode is not possible in a waveguide?

Q.23 Write a short note on "Propagation of waves in a waveguide."